

Using Protocol to create & analyse electrospinning data-set

Introduction:

Electrospinning is an innovative technology used to produce nanofibers with unique properties, ideal for various applications such as drug delivery, filtration, or electronics. However, the quality of the deposit heavily depends on many parameters (voltage, distance, flow rate, environment). The goal of this protocol is to explain how to generate your own dataset to score the quality of the collected deposit. Ultimately, this data will help predict the efficiency of the chosen settings and easily determine whether a combination of parameters is viable or not.

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1-Data-set creation

For create data-set, you need to choose what factors do you want to change and affect values to each factor. For example, I choose 3 factors which 3 values by factor ($3^3 = 27$ possibilities). You don't need to type all the possibilities on you excel table; you can use Box-Behnken design which will create for your excel document with all combinations possible!

<https://numiqo.com/statistics-calculator/design-of-experiments/box-behnken-design-calculator>

- You can tick the Full Factorial-General design and then complete the table with the factor & values.

Screening design
You have many possible factors and little knowledge.

☐ Fractional Factorial Design

☐ Plackett-Burman Design

Full factorial design
You have only a few factors and want the full picture.

☐ Full Factorial - 2-Level

☒ Full Factorial - General

Optimization design
You already know the important factors and want to optimize.

☐ Box-Behnken Design

☐ Central Composite Design

Robust design
You already know the important factors and want stable settings that are less sensitive to noise.

☐ Taguchi Design

Mixture design
Your factors are ingredients or proportions.

☐ Simplex-Centroid

☐ Simplex-Lattice

☐ Extreme-Vertices

Optimal design
You have constraints or no standard design fits your situation.

☐ D-Optimal Design

Factors

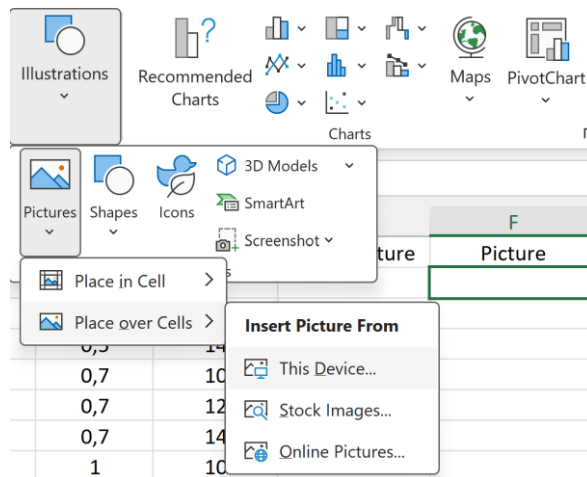
No. of Factors: No. of Replicates: No. of Blocks: No. of Center Points:

Factor	Levels	Level Values
Distance	3	15 18 20
Flow-rate	3	0.5 0.7 1
Voltage	3	10 12 14

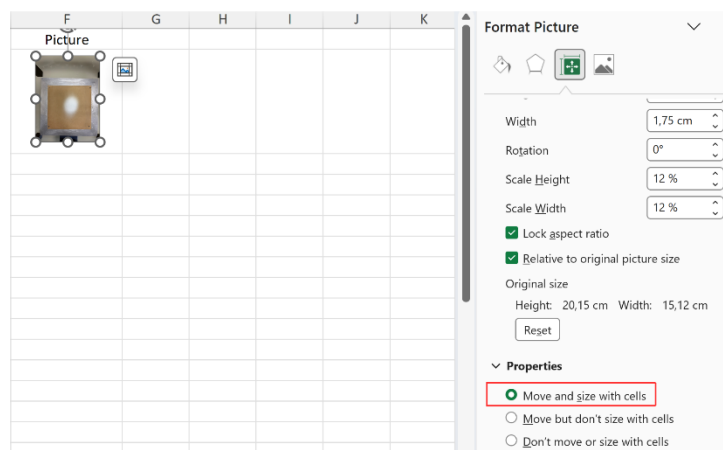
- Then you can have an overview of the Plan and export on excel
- Open excel file and add variable parameter(s) like temperature, pressure... and Picture column

A	B	C	D	E	F
Run Order	Distance	Flow-rate	Voltage	Temperature	Picture
1	15	0,5	10		
2	15	0,5	12		
3	15	0,5	14		
4	15	0,7	10		
5	15	0,7	12		
6	15	0,7	14		
7	15	1	10		

- Start experiments and insert Picture of the collector like the method after :
Illustration → Pictures → **Place over Cells** because if you put the picture in Cell



- Extend the box and reduce the picture size so that it does not extend beyond the frame
- Make right click on the picture and click on **Size & Properties** → **Properties** → **Move and size with cells**



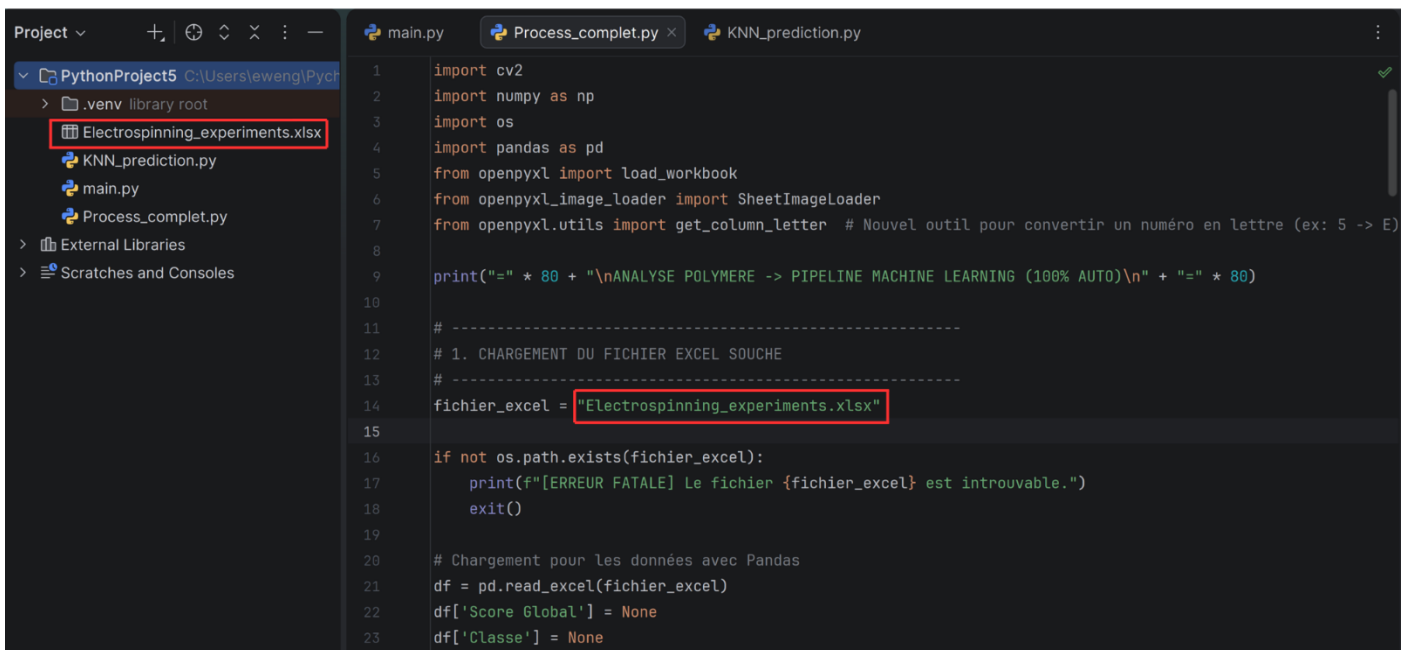
- Follow this process with all combinations

2-Analyse of collector → Classification

When all experiments are finished, it is essential to rate each picture of the collector to see if the deposit quality is good or not. To do this, we will use a Python program that assigns a score to each picture: the lower the score, the better the quality. The program analyses the circularity and texture of the nanofiber deposit. The Python program is available at the following GitHub link:

LIEN GITHUB.

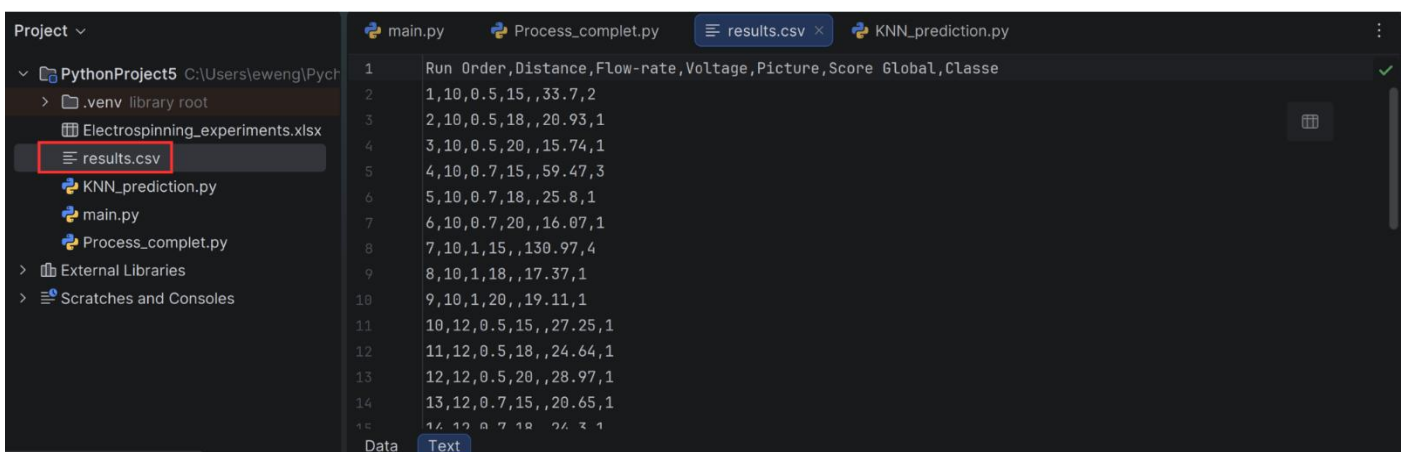
- Insert de Excel table on the lateral window and write the file name on the line 14.



```

1  import cv2
2  import numpy as np
3  import os
4  import pandas as pd
5  from openpyxl import load_workbook
6  from openpyxl_image_loader import SheetImageLoader
7  from openpyxl.utils import get_column_letter # Nouvel outil pour convertir un numéro en lettre (ex: 5 -> E)
8
9  print("=" * 80 + "\nANALYSE POLYMERE -> PIPELINE MACHINE LEARNING (100% AUTO)\n" + "=" * 80)
10
11  # -----
12  # 1. CHARGEMENT DU FICHIER EXCEL SOUCHE
13  # -----
14  fichier_excel = "Electrospinning_experiments.xlsx"
15
16  if not os.path.exists(fichier_excel):
17      print(f"[ERREUR FATALE] Le fichier {fichier_excel} est introuvable.")
18      exit()
19
20  # Chargement pour les données avec Pandas
21  df = pd.read_excel(fichier_excel)
22  df['Score Global'] = None
23  df['Classe'] = None
  
```

- Install the different library if it's necessary
- **Start** the program
- Check if .csv file appear in the lateral window



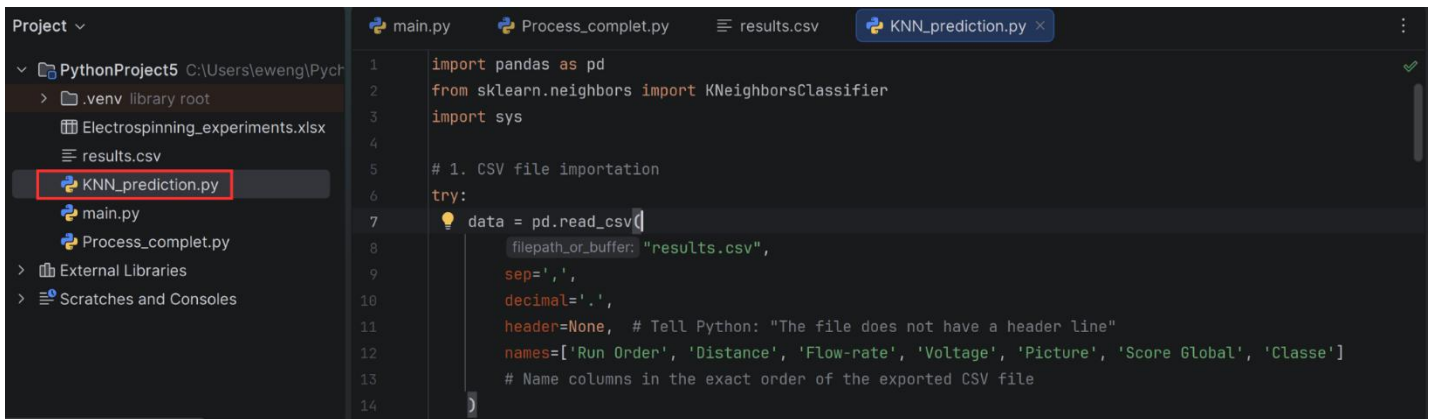
```

1  Run,Order,Distance,Flow-rate,Voltage,Picture,Score Global,Classe
2  1,10,0.5,15,,33.7,2
3  2,10,0.5,18,,20.93,1
4  3,10,0.5,20,,15.74,1
5  4,10,0.7,15,,59.47,3
6  5,10,0.7,18,,25.8,1
7  6,10,0.7,20,,16.07,1
8  7,10,1,15,,130.97,4
9  8,10,1,18,,17.37,1
10 9,10,1,20,,19.11,1
11 10,12,0.5,15,,27.25,1
12 11,12,0.5,18,,24.64,1
13 12,12,0.5,20,,28.97,1
14 13,12,0.7,15,,20.65,1
15 14,12,0.7,18,,24.3,1
  
```

3-Results prediction

In this section, we will see how the trained KNN model can predict results for new parameter combinations, saving you from having to perform additional experiments.

- Open KNN_prediction python file



```

1 import pandas as pd
2 from sklearn.neighbors import KNeighborsClassifier
3 import sys
4
5 # 1. CSV file importation
6 try:
7     data = pd.read_csv(
8         filepath_or_buffer: "results.csv",
9         sep=',',
10        decimal='.',
11        header=None, # Tell Python: "The file does not have a header line"
12        names=['Run Order', 'Distance', 'Flow-rate', 'Voltage', 'Picture', 'Score Global', 'Classe']
13        # Name columns in the exact order of the exported CSV file
14    )

```

- Modify lines 9/10/12 → for the line 11, write the same order like in the csv file!
- Adapt line 35 and 42 if you apply more factors (ex: needle size)
- Adapt the K number line 46 → **odd number** (key → k=3 for 27 samples)
- Choose new values to predict the class line 51 (by defaults 16cm, 0.7mL/h, 15kV)
- **Start** the programme